

Cloud Computing Complete Notes

AWS S3

- Object storage service from AWS.
- Stores data as objects inside buckets.
- Highly scalable and durable.
- Supports lifecycle management and versioning.

Azure Blob Storage

- Microsoft Azure object storage service.
- Stores unstructured data like images and videos.
- Supports hot, cool, and archive tiers.

AWS EC2

- Elastic Compute Cloud provides virtual machines.
- Supports scalable cloud infrastructure.
- Users can choose CPU, RAM, storage, and OS.

Azure Virtual Machines

- Virtual server service from Azure.
- Supports Windows and Linux VMs.
- Used for enterprise applications and hosting.

AWS SageMaker

- Managed machine learning platform.
- Supports training, deployment, and monitoring.
- Provides notebooks, AutoML, and MLOps.

Azure Machine Learning

- Azure AI/ML development platform.
- Supports pipelines, experiments, and deployment.
- Used for enterprise AI workflows.

Vertex AI

- Google Cloud AI platform.
- Supports AutoML, pipelines, and generative AI.
- Provides TPU-based training support.

Docker

- Containerization platform.
- Packages applications with dependencies.
- Uses images and containers.

Orchestration

- Automated coordination of services and containers.
- Handles scaling, deployment, and monitoring.
- Kubernetes is the most popular orchestrator.

Kubernetes

- Container orchestration platform.
- Manages pods, deployments, and services.
- Supports auto scaling and self-healing.

Cloud-Native

- Applications designed specifically for cloud environments.
- Uses containers, microservices, and CI/CD.
- Highly scalable and resilient.

AWS Lambda

- Serverless compute service from AWS.
- Executes code on-demand using events.
- Automatically scales.

Azure Functions

- Serverless computing service from Azure.

- Uses triggers and bindings.
- Ideal for APIs and automation.

Important Concepts

Concept	Meaning
Containers	Lightweight isolated runtime environments
Microservices	Small independent services
Serverless	Run code without managing servers
CI/CD	Continuous Integration and Deployment
Auto Scaling	Automatic increase/decrease of resources
MLOps	Machine Learning Operations
IaC	Infrastructure as Code

End of Notes